Knowledge Based Engineering (KBE) AE4204

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Lecture 1

KBE and the Artificial Intelligence roots

- Definitions

- Expert systems

- Frame Based Systems (part 1)

Contents

- A definition of Knowledge Based Engineering (KBE)

- Knowledge Based Systems (KBS)
  - Working principles: *production rules* and *inference mechanisms*

- Frame Based Systems (FBS)
  - The Object-Oriented paradigm applied to KBS

Intro to Object-Oriented modeling concepts

Knowledge Based Engineering (KBE)

What is it? A methodology? A technology?

Knowledge Based Engineering (**KBE**) is a **methodology** that formalizes, captures, and reuses engineering knowledge to automate and support design and analysis processes.

It is implemented through specific **computational technologies**, referred to as KBE systems, which enable the formalization and capture of, among others:

- Design intent
- Decision logic
- Parametric relations
- Constraints and rules
- Best practices and heuristics
- Process knowledge (including but not limited to geometry manipulation)

Knowledge Based Engineering (KBE)

How does KBE manifest? What does it do in practice?

KBE manifests through the generation of **domain-specific** software applications (**KBE apps**) that use **explicitly encoded** engineering knowledge to automate and support design and analysis tasks.

Key benefit of KBE applications:

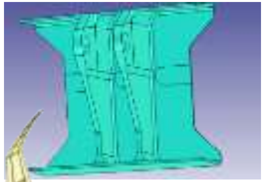
- Automation of repetitive and rule-based design tasks
- Multidisciplinary integration in all phases of the design process
- Formalization and preservation of valuable domain specific knowledge for re-use

KBE: something new?

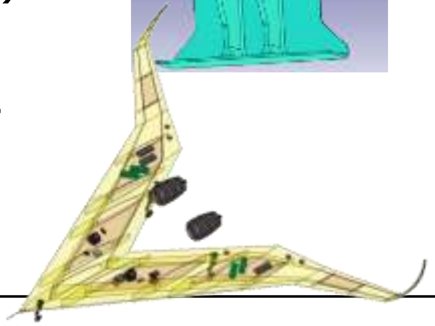
KBE is **NOT** some "new age" way of doing engineering! *....Engineering is ALWAYS based on the use of knowledge!*

KBE is a mainstream technology (>30years) used by engineering companies* to **enhance the level of automation in their product design process**.

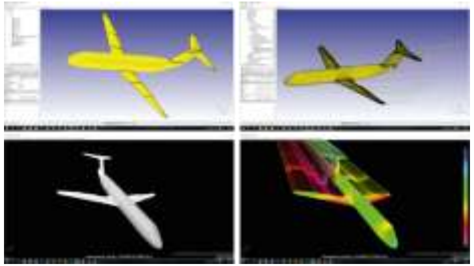
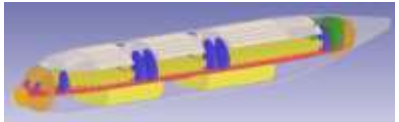
- The most common exploitation of KBE to date concerns with the **automation of detail design** of (structural) components.

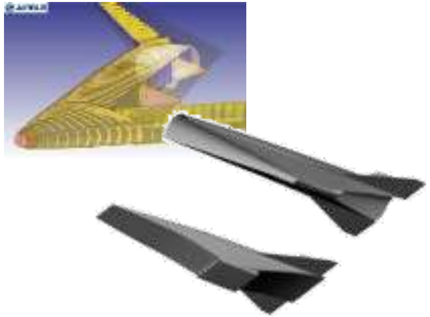
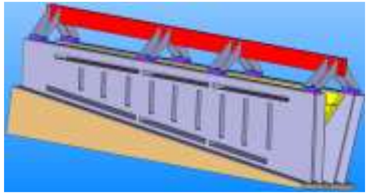


- Exploitation of KBE to support **conceptual design** of complete product configurations is slowly becoming state of the art

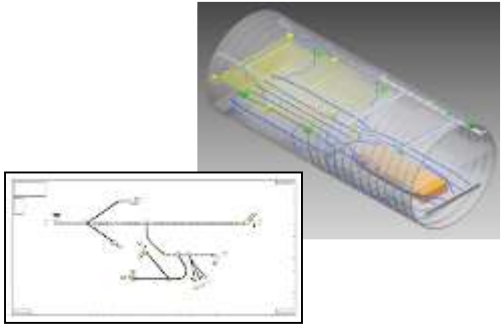
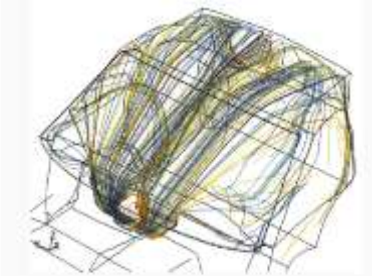
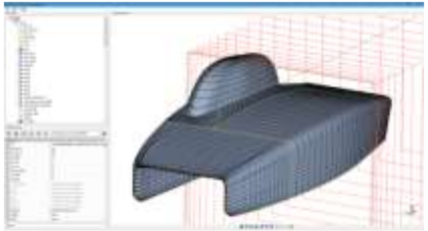
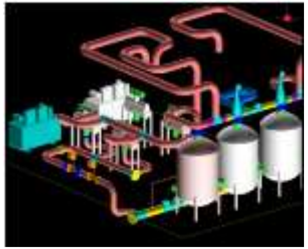


KBE application domains

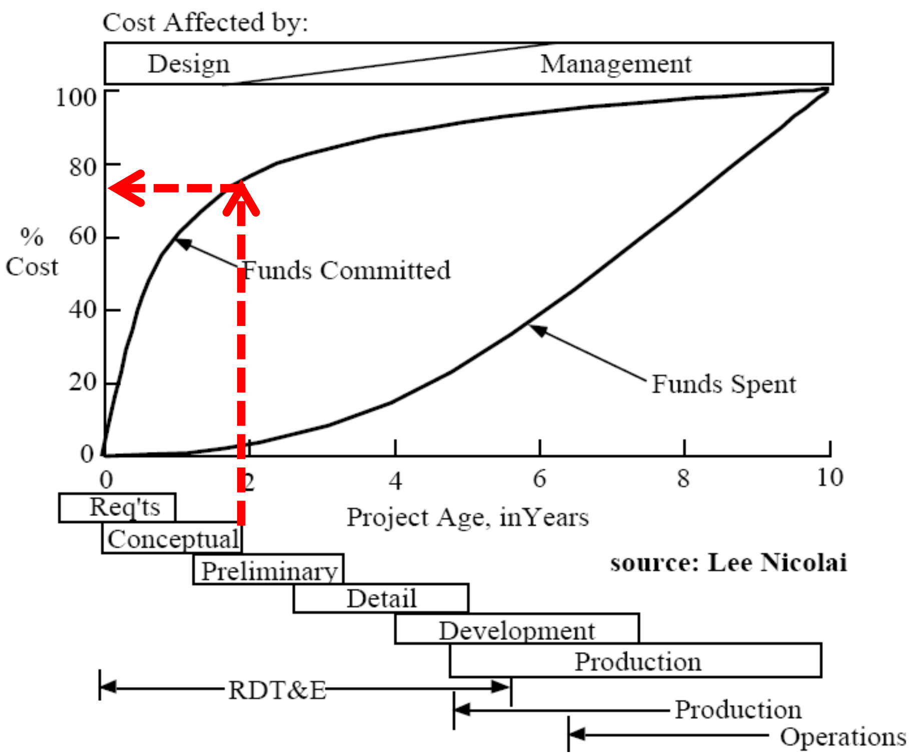



…and so much more!

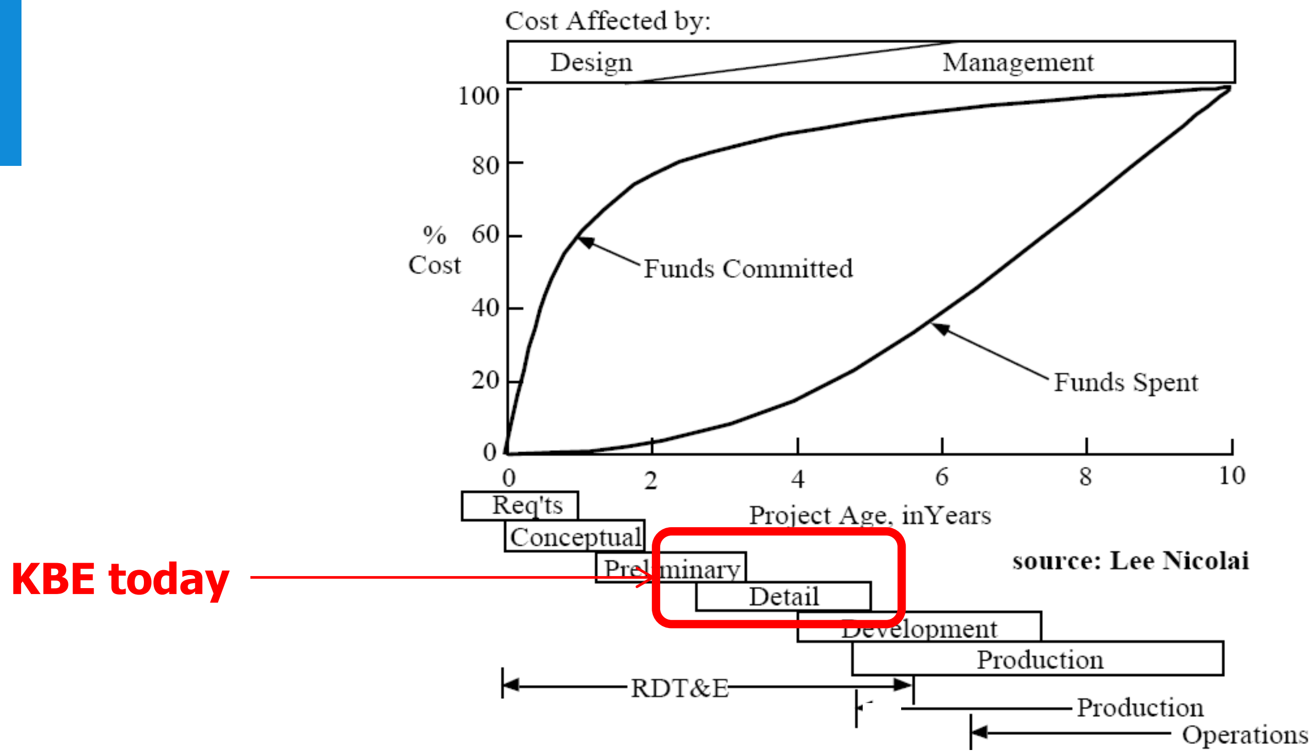





KBE in the product development process



source: Lee Nicolai

KBE in the product development process

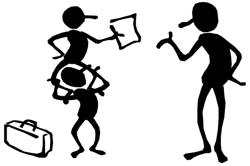


Main phases of the design process

Conceptual design:
- Definition of the performance goals
- Generation and evaluation of competing concepts
- Selection of baseline concept

Time: Weeks→ months

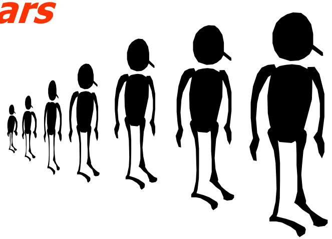
1 %



Preliminary design:
- Refined sizing of the baseline design concept
- Parametric studies
- Global design frozen with the possibility to change only few details

Time: months→ months/years

9 %

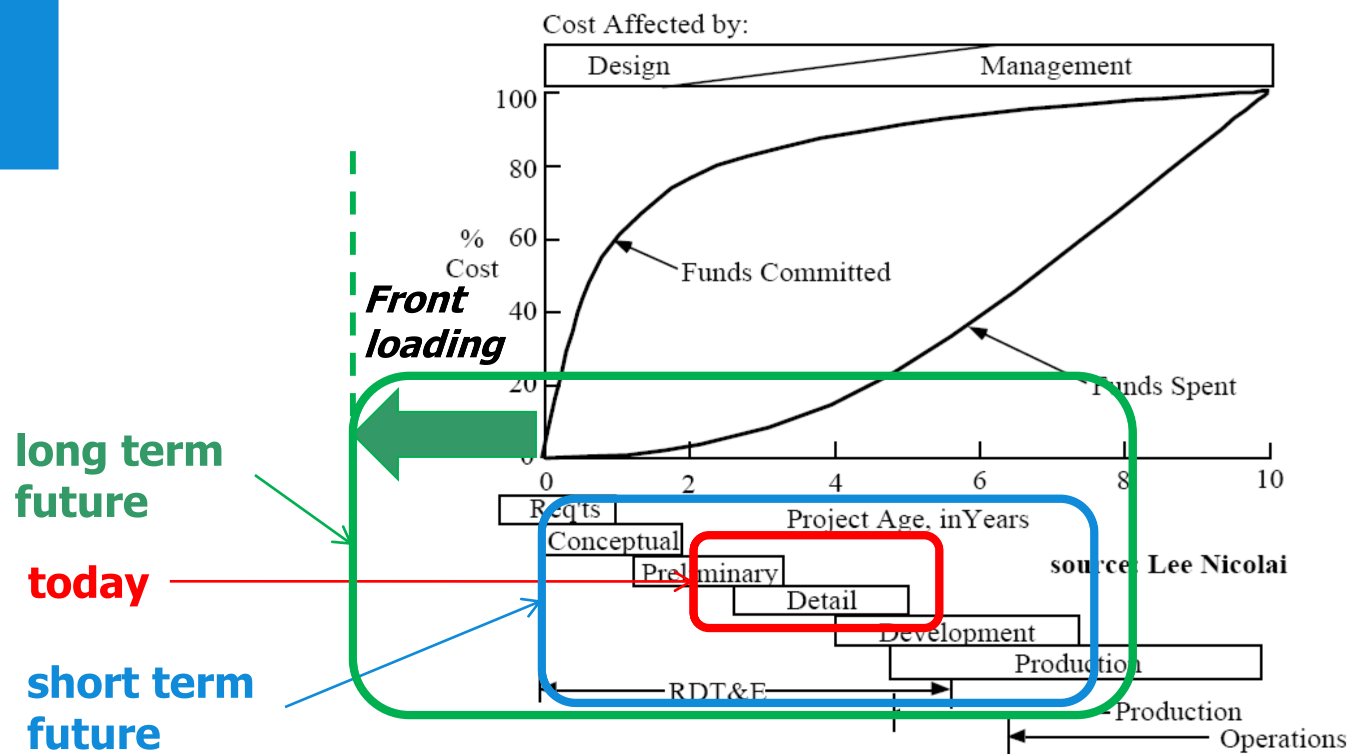


Detail design:
- Design of the product down to the single details
- Accurate evaluation of performances
- Fine tuning of the design
- Release of drawings for production

Time: years

90 %

KBE in the product development process



The AI origin of KBE

The Artificial Intelligence (AI) roots of KBE:

Knowledge Based Systems

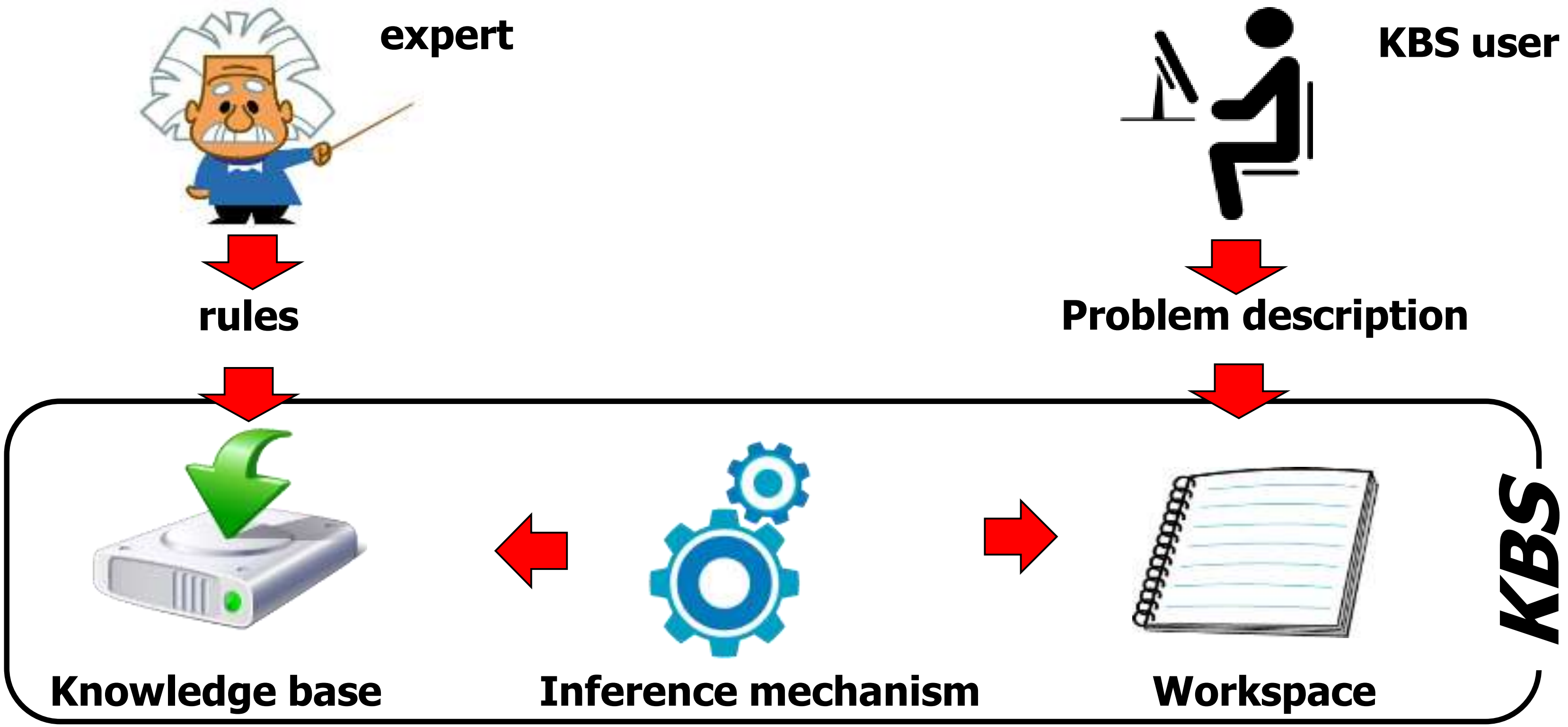
KBE has evolved from the world of **Artificial Intelligence** (AI), from the domain of **Knowledge Based Systems (KBS)**

KBSs are computer applications that use stored knowledge for solving problems in a specific domain

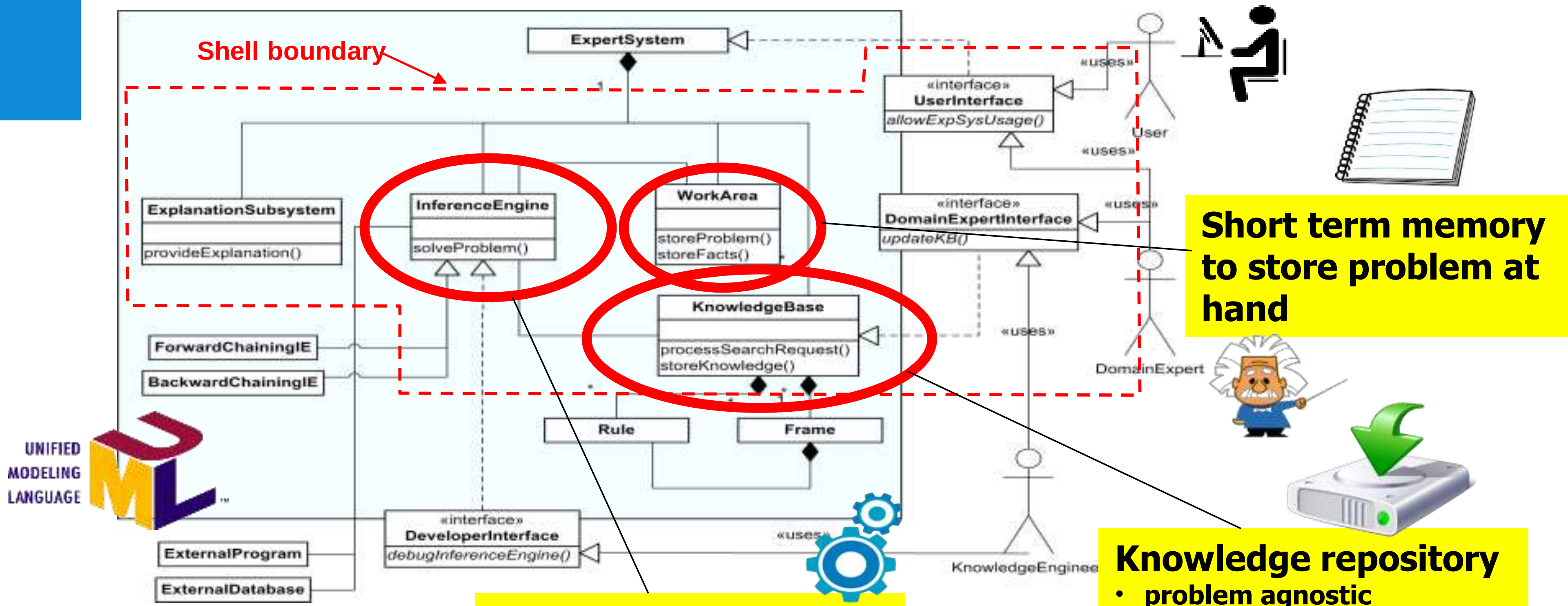
Similarly to a human expert*, a KBS applies **explicit reasoning and logical inference** over stored knowledge to derive answers to posed problems*

Thereby, KBS are also addressed as **expert systems**

KB Systems. Functionality and structure



KB Systems. Functionality and structure



Short term memory to store problem at hand

Reasoning mechanism

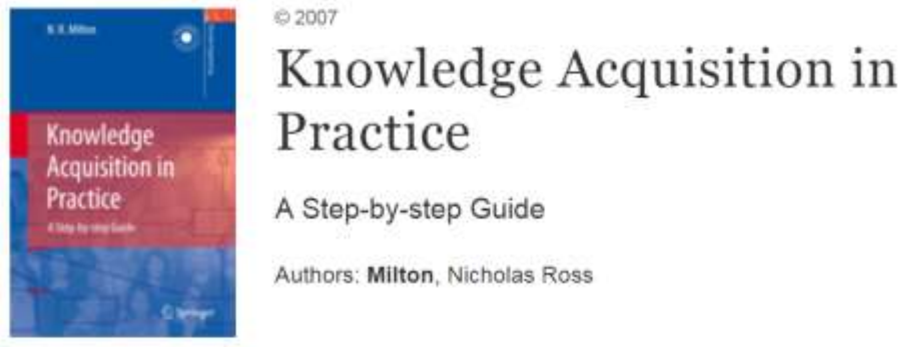
Knowledge repository
- **problem agnostic**
- **to fill with domain specific knowledge**

KB Systems. Functionality and structure

A knowledge base system with an **empty** knowledge base (KB) is just…an empty **shell**!
This shell is what you "buy" when purchasing a KBS
Filling the KB with knowledge is responsibility of the KBS "customer"

How does knowledge get into the KB?

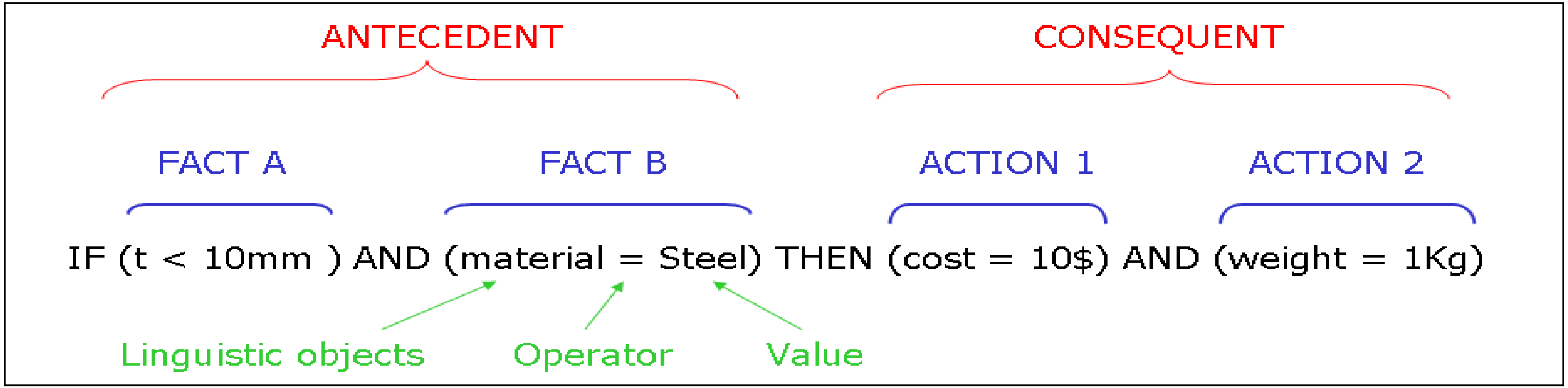
1. You need a domain expert who owns the knowledge
2. You need someone able to elicit the knowledge from the expert (using knowledge acquisition techniques)
3. You need an interface system
4. You need a **symbolic repres**
   that it can be interpreted and



Knowledge representation in KB Systems

Production rules (or simply *rules*) are one of the two most common **symbolic representations** of knowledge for KBS systems.

By rule, the following well know **IF-THEN** construct is intended:



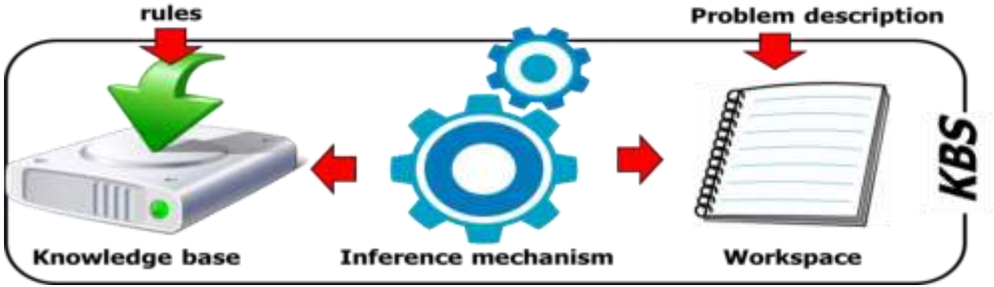
KBSs using exclusively (IF-THEN) rules are called **Rule Based Systems**

The AI origin of KBE

- **Rule based systems**

Inference mechanisms in KB systems

Once the knowledge base has been populated with rules, the inference engine can put them in use for solving problems. ***How?***

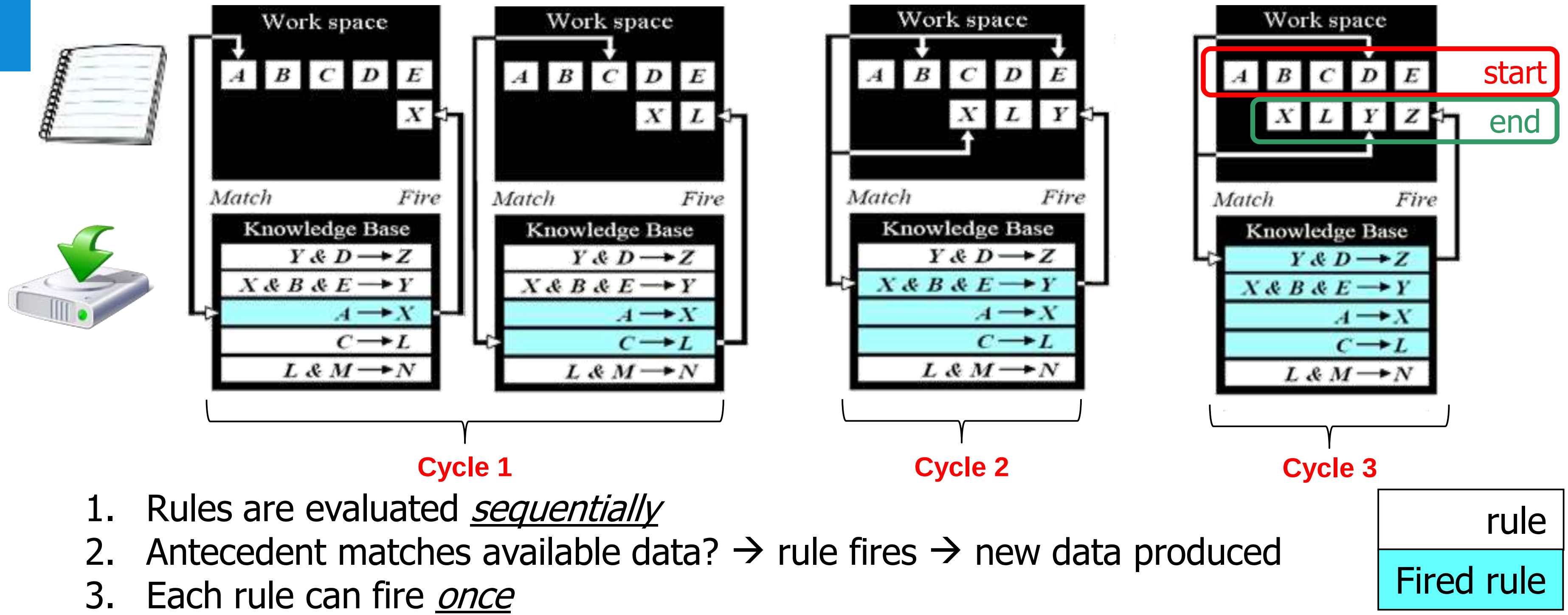


Two main types of inference (i.e. reasoning) mechanisms exist:

1. One based on **forward chaining**: **data* driven** or ***eager approach***
2. One based on **backward chaining**: **goal driven** or "***lazy approach*"***

Inference mechanisms in KB systems

Forward chaining inference technique example:



1. Rules are evaluated *sequentially*
2. Antecedent matches available data? → rule fires → new data produced
3. Each rule can fire *once*

Inference mechanisms in KB systems

Forward chaining (i.e. *data driven* or ***eager*** approach):

on the basis of the data initially available in the KBS work space, the inference engine sequentially and iteratively searches and **fires all fireable** rules in the KB.

Appropriate to perform **analysis and interpretation work**, i.e. to infer *all possible facts* based on available knowledge and initially available facts

It mimics the **behaviour of an expert** who is requested to evaluate a complete scenario, given a set of facts.

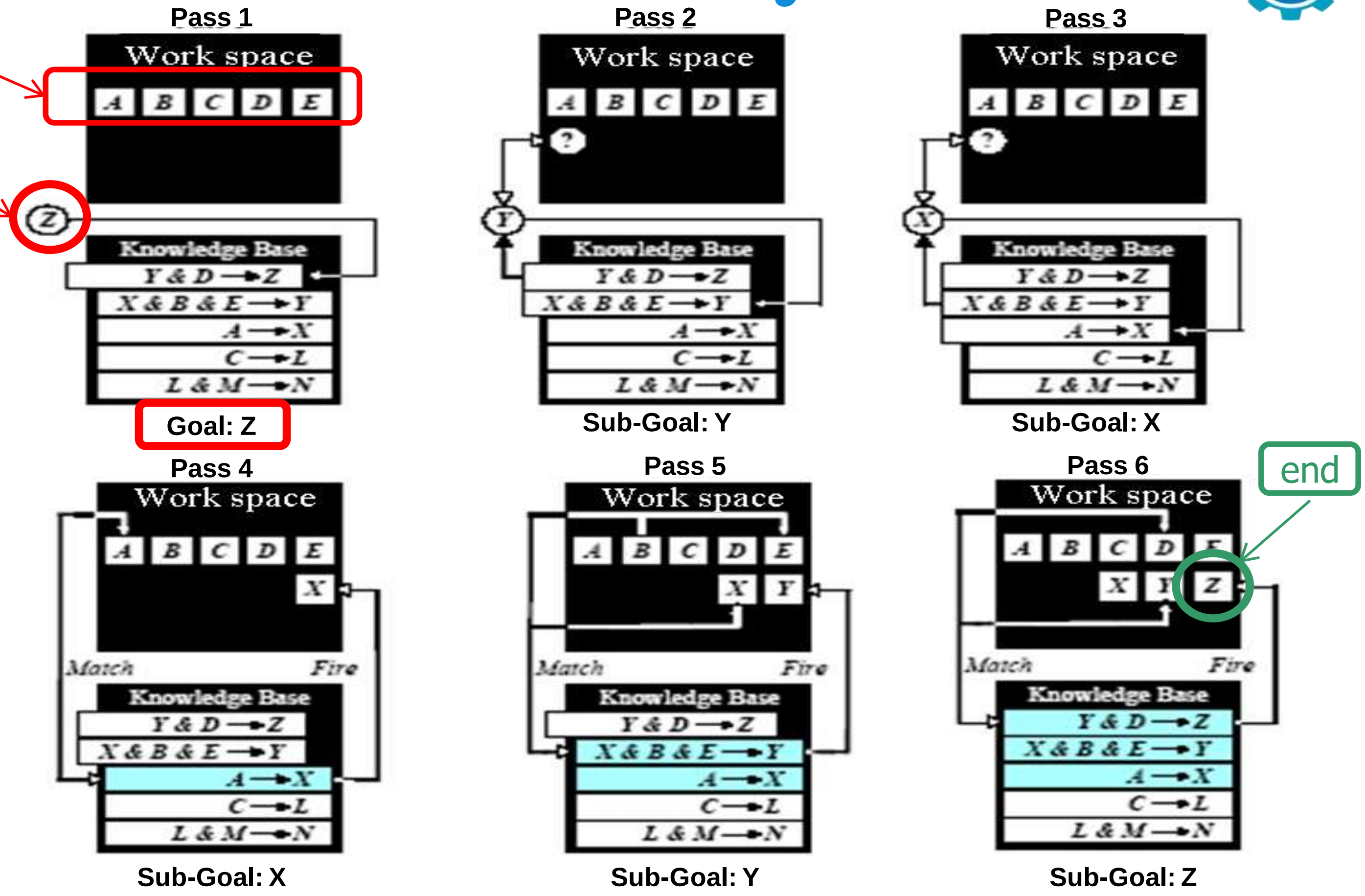
Systematic and **exhaustive,** but **inefficient**

Inference mechanisms in KB systems

start

Hypothesis:
solution is Z

Backward chaining inference technique example:



Stacked rule

Rule

Fired rule

Inference mechanisms in KB systems

Backward chaining (i.e. *goal-driven* approach):

the reasoning process starts with a hypothetical solution (the *goal*) and **evaluates only the rules necessary** to prove it

It mimics the **behaviour of an expert**, who **first** hypothesizes a possible solution to the problem at hand, **and then** looks for evidence to confirm hypothesis.

When facts are missing, the expert asks the customer to help. If facts to prove the hypothesis cannot be found, the expert will switch to a different hypothesis and start all over.

Computationally efficient

[Spoiler alert]: very important for KBE systems

The AI origin of KBE

- **Frame based systems**

Frame Based Systems

Frame Based Systems are another type of KBSs, based on ***frames***:
A frame is a more sophisticated knowledge representation, than just IF-THEN rules

How can I capture the concept of a Fokker 100?

Fokker 100	
MTOW	43000 Kg
No of passengers	107
Wing span	28 m
Total length	35.5 m
Wing area	93.5 m^2
Range	2500 Km

Frame

Frame Based Systems

What about an A380 frame?

Frame name →

Attributes (or *slots*) describing the given object →

← Attributes **values**

	Fokker 100
MTOW	43000 Kg
No of passengers	107
Wing span	28 m
Total length	35.5 m
Wing area	93.5 m^2
Range	2500 Km

Frame Based Systems

Conceptually not too different. Can I abstract?

What about an A380 frame?

Fokker100	
MTOW	43000 Kg
No of passengers	107
Wing span	28 m
Total length	35.5 m
Wing area	93.5 m^2
Range	2500 Km

AirbusA380	
MTOW	560000 Kg
No of passengers	853
Wing span	79,8 m
Total length	73 m
Wing area	845 m^2
Range	15200 Km

Frame Based Systems

Class-frame

Aircraft	
MTOW	Kg
No of pa ssengers	
Wing span	m
Total length	m
Wing area	m^2
Range	Km

Fokker100: Aircraft	
MTOW	43000 Kg
No of passengers	107
Wing span	28 m
Total length	35.5 m
Wing area	93.5 m^2
Range	2500 Km

different **instance-frames**
of the same class-frame

AirbusA380: Aircraft	
MTOW	560000 Kg
No of passengers	853
Wing span	79,8 m
Total length	73 m
Wing area	845 m^2
Range	15200 Km

Frame Based Systems

A **Class-frame** is used as template to define **a type of objects**
E.g. the class-frame *Aircraft* can generate the Fokker100, the A380, the B737, etc…which are all different objects, all of type **Aircraft**

Instance-frames are used to define **specific instances (objects)** of a generic class. E.g. Fokker100 is just one of the many possible instances of the generic class frame *Aircraft*

Instance-frames are generated by assigning a set of specific <u>values</u> to the attributes of a class-frame.

Any given set of attribute <u>values</u> yields a <u>different</u>, <u>specific and unique</u> instance-frame of the <u>same</u> class-frame.

Frame Based Systems

The object-oriented paradigm applied to knowledge based systems.

The concepts of **class**-frames and **instance**-frames are not different from the concepts of **classes** and **objects** in the **Object Oriented (OO) paradigm***.

According to the OO approach to model knowledge, each entity (or object) is a **unique** instantiation of a generic class.

A class is a model for a **type of** objects. It is the **blueprint**, or **template**, or **factory** from which individual objects are created.

Frame Based Systems

The object-oriented paradigm applied to knowledge based systems.

class

instances

Frame Based Systems

The object-oriented paradigm applied to knowledge based systems.

Mammal	
Has skeleton	
Produces milk	
Has neocortex	
Can fly	
Lays egg	

The five slots are sufficient to define what a mammal is, but not to describe what mammals are. But, luckily the OO paradigm allow us to define more structured class hierarchies…..

Main reference material for this lecture

- **Chapter 3** *Knowledge Based Engineering. The AI roots and the OO paradigm*

- **Journal paper** *(G. La Rocca, Knowledge based engineering: Between AI and CAD. Review of a language based technology to support engineering design, Advanced Engineering Informatics, 26 (2012) pp159–179*

All available in the section *Reading Material* on